

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I ____R. AKSHAYA(192524118)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _R. Akshaya_____

Assessment Tasks

Analytical Problem Solving

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.
Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
4. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be

configured.

Discuss monitoring and alerting strategies for proactive issue detection.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Assessment Tasks

Analytical Problem Solving

1. A media company wants to migrate its video streaming platform to the cloud.

The system must handle millions of concurrent users with minimal buffering.

Design a cloud architecture using scalability and caching techniques.

Explain how load balancing and auto-scaling can ensure smooth streaming.

Discuss cost optimization strategies for such high-demand workloads.

Cloud Architecture using Scalability and Caching

- Deploy the application across multiple cloud regions to ensure high availability and reduce latency for users worldwide.
- Store video content in scalable cloud object storage that can handle large volumes of media files without performance degradation.
- Use a Content Delivery Network (CDN) to cache popular videos at edge locations, allowing users to stream content from nearby servers.
- Deploy application servers behind load balancers so incoming requests are evenly distributed across multiple instances.
- Use distributed databases with read replicas to efficiently manage user profiles, subscriptions, and streaming history.

Load Balancing and Auto-Scaling

- Load balancers distribute incoming traffic evenly across available servers, preventing any single server from becoming overloaded.
- Auto-scaling automatically launches additional server instances during peak viewing hours and removes them during low traffic periods.
- Health monitoring detects unhealthy servers and automatically redirects traffic to healthy instances.
- Multiple availability zones ensure uninterrupted streaming even if one data center experiences failure.
- This combination minimizes buffering, improves response time, and provides a seamless viewing experience.

Cost Optimization Strategies

- Use auto-scaling to pay only for the resources required based on current demand.
- Store frequently accessed videos in CDN caches while moving rarely accessed videos to lower-cost archival storage.
- Use reserved instances for predictable workloads and spot instances for non-critical background tasks such as video encoding.

- Monitor cloud resource usage regularly to identify and eliminate unused resources.
- Compress video files efficiently to reduce storage and bandwidth costs without significantly affecting quality.

2. A fintech startup is deploying applications using containers across multiple cloud providers.

They face challenges in maintaining consistency and managing deployments.

Propose a solution using container orchestration and multi-cloud strategies.

Explain how service discovery and scaling can be handled efficiently.

Highlight the benefits and risks of a multi-cloud deployment model.

Container Orchestration and Multi-Cloud Strategy

- Package applications using Docker containers to ensure consistent execution across different cloud platforms.
- Deploy and manage containers using Kubernetes, which automates scheduling, deployment, scaling, and recovery.
- Use Infrastructure as Code (IaC) tools like Terraform to create identical cloud environments across providers.
- Maintain centralized configuration management to ensure consistency in application settings.
- Implement CI/CD pipelines to automate testing and deployment across all cloud environments.

Service Discovery and Scaling

- Kubernetes service discovery automatically enables containers to locate and communicate with each other using internal DNS.
- Horizontal Pod Autoscaler increases or decreases container instances based on CPU, memory, or custom metrics.
- Load balancers distribute incoming requests among healthy containers to improve application performance.
- Self-healing features automatically restart failed containers and replace unhealthy pods.
- Monitoring tools continuously track container performance and trigger scaling when required.

Benefits and Risks of Multi-Cloud

Benefits

- Prevents vendor lock-in by allowing applications to run on multiple cloud providers.
- Improves disaster recovery and business continuity if one provider experiences downtime.
- Enables organizations to choose the best cloud services based on performance and cost.
- Enhances availability by distributing workloads across different cloud platforms.
- Provides greater flexibility in meeting regional compliance requirements.

Risks

- Managing multiple cloud platforms increases operational complexity.
- Data transfer between cloud providers may increase network costs.
- Maintaining consistent security policies across clouds can be challenging.
- Monitoring and troubleshooting become more difficult due to different cloud tools.
- Staff require additional skills to manage multiple cloud environments.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.

The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.

Provide steps to ensure business continuity during failures.

Cloud-Native Backup, Replication, and Versioning

- Schedule automatic backups of databases, application files, and user records at regular intervals.
- Replicate critical data to multiple geographic regions to protect against regional failures.
- Enable object storage versioning to recover files that are accidentally deleted or modified.
- Use cloud snapshot services to capture complete system states for quick restoration.
- Regularly verify backup integrity to ensure backups can be restored successfully.

Achieving RTO and RPO

- Recovery Time Objective (RTO) is achieved by maintaining standby infrastructure that can quickly replace failed systems.
- Recovery Point Objective (RPO) is minimized through continuous data replication and frequent incremental backups.
- Automated recovery procedures reduce downtime and speed up service restoration.
- Disaster recovery drills help verify that RTO and RPO targets can be achieved during actual failures.
- Monitoring systems continuously track backup status and replication health.

Business Continuity During Failures

- Maintain backup copies in different cloud regions to avoid data loss caused by regional disasters.
- Implement automated failover mechanisms to switch users to backup systems without manual intervention.
- Regularly test disaster recovery plans and document recovery procedures.
- Train IT teams on emergency response processes to ensure quick recovery.
- Continuously monitor applications, storage, and network infrastructure to detect failures early.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.

Some applications must remain on-premises due to compliance requirements.

Design a hybrid architecture that ensures secure communication between environments.

Explain how identity management and encryption should be implemented.

Discuss challenges in managing hybrid cloud environments.

Hybrid Cloud Architecture

- Keep confidential government applications and sensitive citizen data on secure on-premises infrastructure.
- Deploy non-sensitive applications such as public websites and information portals in the public cloud.
- Connect on-premises infrastructure and cloud environments using secure VPN or dedicated private network connections.

- Use firewalls and network segmentation to isolate sensitive systems from public-facing services.
- Synchronize data securely between environments while complying with government regulations.

Identity Management and Encryption

- Implement centralized Identity and Access Management (IAM) with role-based access control (RBAC).
- Enable Multi-Factor Authentication (MFA) for administrators and users accessing sensitive systems.
- Encrypt data during transmission using TLS/SSL protocols to prevent interception.
- Encrypt stored data using strong encryption algorithms such as AES-256.
- Store encryption keys securely using cloud key management services or hardware security modules.

Challenges in Managing Hybrid Cloud

- Ensuring consistent security policies across cloud and on-premises environments.
- Managing network latency and bandwidth between connected infrastructures.
- Integrating legacy applications with modern cloud-native services.
- Maintaining regulatory compliance across different environments.
- Monitoring, patching, and managing multiple infrastructures increases administrative effort.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.

The system must continue operating even if one region goes down.

Propose a multi-region deployment strategy in the cloud.

Explain how data replication and failover mechanisms should be configured.

Discuss monitoring and alerting strategies for proactive issue detection.

Multi-Region Deployment Strategy

- Deploy the SaaS application across multiple cloud regions using an **active-active** or **active-passive** architecture to ensure continuous availability.
- Place a **global load balancer** in front of all regions to direct user requests to the nearest and healthiest application instance, reducing latency and improving user experience.

- Host application servers, databases, and storage resources in each region so that every region can independently serve users if another region becomes unavailable.
- Distribute workloads across multiple availability zones within each region to protect against data center failures.
- Regularly synchronize application configurations across all regions to maintain consistency and simplify maintenance.

Data Replication and Failover Mechanisms

- Configure **database replication** between regions using synchronous replication for critical transactional data and asynchronous replication for less time-sensitive data.
- Enable **automatic failover**, allowing the system to immediately switch user traffic to a healthy region whenever the primary region experiences an outage.
- Use **DNS-based failover** or a global traffic manager to automatically redirect users to the available region without requiring manual intervention.
- Continuously replicate application files, user data, and configuration settings to ensure that backup regions always contain up-to-date information.
- Perform regular disaster recovery and failover testing to verify that recovery procedures work correctly and downtime is minimized.

Monitoring and Alerting Strategies

- Continuously monitor infrastructure components such as CPU, memory, storage, network bandwidth, database performance, and application response time using cloud monitoring tools.
- Configure **real-time alerts** to notify administrators immediately when resource utilization exceeds thresholds, services become unavailable, or application errors increase.
- Use centralized logging and monitoring dashboards to collect logs from all regions, making it easier to identify, troubleshoot, and resolve issues quickly.
- Implement health checks for servers, databases, and application services so unhealthy resources are automatically removed from the load balancer until they recover.
- Analyze performance metrics and usage trends regularly to perform capacity planning, predict future demand, and optimize cloud resources while maintaining high availability and fault tolerance.